

Experiment: Homogeneous Transformation for Planar and Spatial Mechanisms Using Python Robotics Toolbox

Duration: 2 Hours

GENERAL INSTRUCTIONS

For each assigned task, students are required to:

1. Develop the Python program using the Python Robotics Toolbox.
 2. Generate the required homogeneous transformation matrix.
 3. Display the transformation matrix.
 4. Determine the final position and orientation.
 5. Visualize the coordinate frame wherever applicable.
 6. Record the observations and results.
 7. Submit the Python program and output screenshots.
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PART A – PLANAR MECHANISM (SE2)

TASK 1 – PLANAR TRANSLATION

A planar robotic link is initially located at the origin of the XY coordinate system. The link is translated by **4 units along the X-axis** and **3 units along the Y-axis**.

Student Tasks

- a) Generate the homogeneous transformation matrix using Python Robotics Toolbox.
 - b) Display the transformation matrix.
 - c) Determine the final position of the coordinate frame.
 - d) Plot the original and transformed coordinate frames.
 - e) Repeat the experiment for the following translations:
 - (2, 5)
 - (5, 1)
 - (-3, 4)
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TASK 2 – PLANAR ROTATION

A planar robotic link is initially aligned with the X-axis. The link is rotated counter-clockwise by **60°**.

Student Tasks

- a) Generate the rotation transformation matrix.
 - b) Display the transformation matrix.
 - c) Determine the final orientation of the link.
 - d) Plot the transformed coordinate frame.
 - e) Repeat the experiment for:
 - 30°
 - 90°
 - 120°
 - 180°
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TASK 3 – PLANAR TRANSLATION AND ROTATION

A planar robotic link undergoes the following transformation:

- Translation = (3, 2)
- Rotation = 45°

Student Tasks

- a) Generate the translation transformation.
- b) Generate the rotation transformation.
- c) Combine the two transformations.
- d) Display the final homogeneous transformation matrix.
- e) Determine the final position and orientation.
- f) Plot the final coordinate frame.

Repeat the experiment for:

Case Translation Rotation

1	(3, 2)	45°
2	(5, 1)	60°
3	(2, 4)	90°

TASK 4 – EFFECT OF TRANSFORMATION ORDER

A planar robotic system undergoes:

- Translation = (4, 2)

- Rotation = 90°

Calculate the final transformation for the following two cases.

Case A

Translation followed by Rotation.

Case B

Rotation followed by Translation.

Student Tasks

- Calculate the transformation matrix for Case A.
- Calculate the transformation matrix for Case B.
- Compare the two matrices.
- Compare the final positions.
- Explain why the final results are different.

TASK 5 – SEQUENTIAL PLANAR TRANSFORMATION

A planar robot initially starts at the origin and performs the following sequence:

1. Translate by (2, 1)
2. Rotate by 30°
3. Translate by (3, 2)
4. Rotate by 45°

Student Tasks

- Create the transformation matrix for each operation.
- Combine all transformations in the correct sequence.
- Calculate the final homogeneous transformation matrix.
- Determine the final X and Y coordinates.
- Determine the final orientation.
- Plot the final coordinate frame.
- Change the order of the transformations and compare the results.

PART B – SPATIAL MECHANISM (SE3)

TASK 6 – 3D TRANSLATION

A robotic end-effector is initially located at the origin. It is translated by:

X = 3 units, Y = 4 units, Z = 2 units

Student Tasks

- a) Generate the SE3 transformation matrix.
- b) Display the transformation matrix.
- c) Determine the final position of the end-effector.
- d) Plot the transformed coordinate frame.
- e) Repeat for:

Case X Y Z

1 3 4 2

2 5 2 3

3 -2 4 5

TASK 7 – ROTATION ABOUT X, Y AND Z AXES

A robotic end-effector is initially aligned with the reference coordinate frame.

Perform the following rotations independently:

- 45° about X-axis
- 60° about Y-axis
- 90° about Z-axis

Student Tasks

- a) Generate the transformation matrix for rotation about X-axis.
 - b) Generate the transformation matrix for rotation about Y-axis.
 - c) Generate the transformation matrix for rotation about Z-axis.
 - d) Compare the three transformation matrices.
 - e) Plot the resulting coordinate frames.
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TASK 8 – 3D TRANSLATION FOLLOWED BY ROTATION

A robot end-effector is first translated to the position:

(2, 3, 4)

It is then rotated by **60° about the Z-axis**.

Student Tasks

- a) Generate the translation matrix.
- b) Generate the rotation matrix.
- c) Combine the transformations.
- d) Determine the final homogeneous transformation matrix.
- e) Determine the final position.
- f) Determine the final orientation.
- g) Plot the final coordinate frame.

Repeat the experiment using rotations about:

- X-axis
- Y-axis
- Z-axis

TASK 9 – MULTIPLE ROTATIONS

A robotic tool undergoes the following rotations:

1. 30° about X-axis
2. 45° about Y-axis
3. 60° about Z-axis

Student Tasks

- a) Generate the individual rotation matrices.
- b) Calculate the combined transformation matrix.
- c) Determine the final orientation.
- d) Plot the final coordinate frame.
- e) Repeat the calculation using the reverse order:
Z-axis → Y-axis → X-axis
- f) Compare both results.
- g) Explain why the results are different.

TASK 10 – COMPLETE SPATIAL TRANSFORMATION

A robotic end-effector initially located at the origin undergoes the following sequence:

1. Translation by (2, 3, 1)
2. Rotation by 30° about X-axis

3. Rotation by 45° about Y-axis
4. Rotation by 60° about Z-axis

Student Tasks

- a) Create the transformation matrix for each operation.
 - b) Combine the transformations in the given order.
 - c) Calculate the final homogeneous transformation matrix.
 - d) Determine the final X, Y and Z coordinates.
 - e) Determine the final orientation.
 - f) Visualize the final coordinate frame.
 - g) Change the order of the rotations and compare the final transformation.
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PART C – APPLICATION-BASED TASKS

TASK 11 – TWO-LINK PLANAR ROBOT

Consider a two-link planar robotic arm.

Link 1

- Length = 4 units
- Joint angle = 30°

Link 2

- Length = 3 units
- Joint angle = 45° relative to Link 1

Student Tasks

- a) Construct the transformation matrix for Link 1.
 - b) Construct the transformation matrix for Link 2.
 - c) Calculate the overall transformation matrix.
 - d) Determine the end-effector position.
 - e) Determine the end-effector orientation.
 - f) Plot the two-link planar robot.
 - g) Change the joint angles and observe the change in end-effector position.
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TASK 12 – TWO-LINK SPATIAL ROBOT

Consider a simple 3D robotic arm consisting of two links.

Link 1

- Translation = (0, 0, 3)
- Rotation = 30° about Z-axis

Link 2

- Translation = (2, 0, 0)
- Rotation = 45° about Y-axis

Student Tasks

- a) Construct the transformation matrix for Link 1.
 - b) Construct the transformation matrix for Link 2.
 - c) Calculate the overall transformation matrix.
 - d) Determine the end-effector position.
 - e) Determine the final orientation.
 - f) Plot the coordinate frames.
 - g) Change the rotation angles and observe the effect on the end-effector.
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PART D – CHALLENGE TASK

TASK 13 – COMPLETE TRANSFORMATION SEQUENCE

A robot starts from the world coordinate frame and performs the following sequence.

Planar Transformation

1. Translate by (2, 3)
2. Rotate by 45°
3. Translate by (4, 0)
4. Rotate by 30°

Spatial Transformation

1. Translate by (2, 3, 1)
2. Rotate by 30° about X-axis
3. Rotate by 45° about Y-axis
4. Translate by (1, 2, 3)
5. Rotate by 60° about Z-axis

Student Tasks

- a) Write a Python program to perform all the transformations.

- b) Calculate the final homogeneous transformation matrix.
 - c) Extract the final position from the transformation matrix.
 - d) Determine the final orientation.
 - e) Plot the final coordinate frame.
 - f) Change the order of two or more transformations.
 - g) Compare the original and modified transformation matrices.
 - h) Explain the effect of transformation order on the final position and orientation.
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OBSERVATION TABLE

Task No.	Transformation Type	Final X	Final Y	Final Z	Final Orientation
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